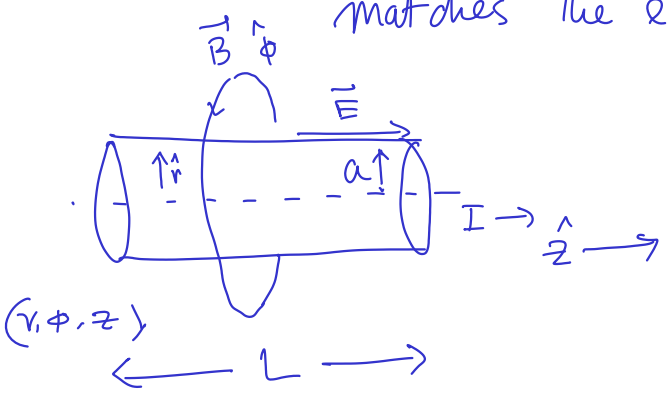
Assignment - 6

Q1 A straight cylindrical wire of radius a, length L and resistance R carries a steady current I.

(a) Calculate electric field $\vec{E}$ and magnetic field $\vec{B}$ at the surface of the wire.

(b) Calculate the Poynting vector $\vec{S}$ at the surface of the wire

(c) Integrate the Poynting vector over the surface of the wire to find the total power flowing into the conductor. Further, verify that it matches the expected Joule heating formula I^2R



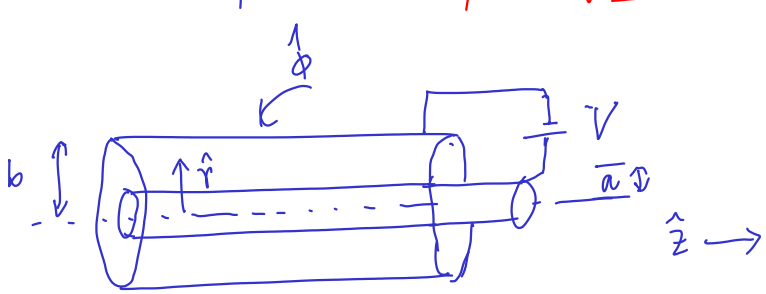
Coord. system (r, ϕ, z)

Q2 A long, straight co-axial cable consists of an inner cylindrical conductor of radius a and a thin outer cylindrical conducting shell of radius b. A DC battery maintains a steady potential difference V between the conductors and a steady current I flows down the inner conductor and returns via the outer conductor. Assume the space between the conductors is vacuum.

(a) Determine the electric field $\vec{E}$ and magnetic field $\vec{B}$ in the region $a < r < b$ where r is the radial distance from the central axis.

(b) Calculate the Poynting vector $\vec{S}$ in the given region

(c) Integrate the poynting vector over the cross-sectional area between the conductors to find the total power P transmitted down the cable and verify that it matches the standard expression $P = VI$.



Coord. system: (r, ϕ, z)